

Practical 4

Name:Pragati Santosh Gaikwad

PRN:202501050136

Course

nitzeo.codedtantra.com/secure/course.php?courseid=699c548c3dab96214c8f97d2e770e0e0a588c54804eb396214c3b7d50

CodeTANTRA Home Learn Anywhere

202301040195@nitzeo.ac.in Support Logout

Essentials of Data Science Laboratory - 2304102L - 2026

Search course

- 1. Practical 1
- 2. Practical 2
- 3. Practical 3**
- 4. Practical 4
- 5. Practical 5

Practical 3

About this unit

Practical 3

Practice Lab Assignment

Unit - 100% completed

Lab Assignment

Unit - 100% completed

The screenshot displays the CodeTANTRA web interface. On the left, a sidebar lists course sections: 'Essentials of Data Science Laboratory - 2304102L - 2026', with sub-items like '1. Practical 1', '2. Practical 2', '3. Practical 3', '4. Practical 4', '4.1. Practice Lab Assignment', and '4.1.1. Pandas - series creation and manipu...'. The main content area is titled '4.1.1. Pandas - series creation and manipulation' and contains instructions to write a Python program that takes a list of numbers from the user, creates a Pandas series, and calculates the mean of even and odd numbers separately using `groupby` and `mean()` operations. It also provides a hint about grouping series based on even/odd status and specifies input/output formats. A 'Sample Test Cases' button is visible. On the right, a code editor shows the solution code, which uses `input()` to get a list of numbers, creates a Pandas series, groups it by even/odd status, and prints the mean for each group. Below the code, a test runner indicates '3 out of 3 shown test case(s) passed' and '3 out of 3 hidden test case(s) passed'. It also shows a comparison between 'Expected output' and 'Actual output', both displaying the same results for a sample input.

Course

mitaoe.codetantra.com/secure/course.jsp?eucld=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d5e/698c54860aba96214c8b7d5e/667aa307282204631649bb62

CodeTantra Teach &... Adobe Acrobat Course

CODETANTRA

Home Learn Anywhere

202501040136@mitaoe.ac.in Support Logout

4.1.2. Dictionary to dataframe

1311

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

datafram...

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

```
1 # import pandas as pd
2
3 ## Provided dictionary of lists
4 #data = {
5 #    'Name': ['Alice', 'Bob', 'Charlie'],
6 #    'Age': [25, 30, 35]
7 #}
8
9 ## Convert the dictionary to a DataFrame
10 #df = pd.DataFrame(data)
11
12 ## Display the original DataFrame
13 #print("Original DataFrame:")
14 #print(df)
15
16 # Add a new row
```

Average time

0.106 s

106.50 ms

Maximum time

0.130 s

130.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 (130 ms)

Debug

Expected output

Original DataFrame:

..... Name Age

0..... Alice...25

1..... Bob...30

2..... Charlie...35

New name: Susan

Actual output

Original DataFrame:

..... Name Age

0..... Alice... 25

1..... Bob...30

2..... Charlie...35

New name: Susan

Terminal

Test cases

36°C

Clear

Search

ENG IN

22:37

03-05-2025

Course

mitaoe.codetantra.com/secure/course.jsp?eucld=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d5e/698c54860aba96214c8b7d5e/667aa307282204631649bb62

CodeTantra Teach &... Adobe Acrobat Course

CODETANTRA

Home Learn Anywhere

202501040136@mitaoe.ac.in Support Logout

4.1.3. Student Information

1312

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade to be 'B').

Notes:

Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

1

2

3

4

5

6

7

8

9

10

11

12

13

14

```
1 # import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep='\\', header=None, names=['Name', 'Age', 'Grade'])
6 print("First five rows:")
7 print(data.head())
8 data['Age'].round()
9 av_age = data['Age'].mean().2)
10 print("Average age: ",av)
11 fil_data = data[data['Grade'].isin(['A', 'B'])]
12 print("Students with a grade up to B")
13 print(fil_data)
14
```

Average time

0.061 s

61.50 ms

Maximum time

0.085 s

85.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 (85 ms)

Debug

Expected output

studentdata.txt

First five rows:

..... Name Age Grade

0..... Alice 25 A

1..... Bob 30 B

2..... Charlie 35 A

Actual output

studentdata.txt

First five rows:

..... Name Age Grade

0..... Alice 25 A

1..... Bob 30 B

2..... Charlie 35 A

Terminal

Test cases

36°C

Clear

Search

ENG IN

22:38

03-05-2025

4.2.3. City that Sold the Most Products

Write a Python program that takes the filename of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by city and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

```

Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,6,20,Los Angeles
2025-01-04,Product C,5,30,New York
2025-01-04,Product B,4,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,6,30,Los Angeles

```

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

```

1 # Import pandas as pd
2 # Prompt the user for the file name
3 file_name = input()
4
5 # Load the data
6 df = pd.read_csv(file_name)
7
8 city_df = df.groupby('City')['Quantity'].sum()
9 best_city = city_df.idxmax()
10
11 # Display the result
12 print(f"City sold the most products: {best_city}")

```

Test Case 1: City sold the most products: Los Angeles

4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using `info()`).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
- Drop the 'Cabin' column due to many missing values.
- Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below:

PassengerId	Survived	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	Mr. Bruce Ismay	Male	29	0	0	15167	51.0	C85	S
2	1	Mr. Mr. William	Male	34	1	0	17463	51.0	C123	S
3	1	Mr. John Bradley	Male	22	1	0	17599	51.0	C123	S
4	1	Mr. John Bradley	Male	22	1	0	17599	51.0	C123	S
5	1	Mr. John Bradley	Male	22	1	0	17599	51.0	C123	S

Sample Data:

```

PassengerId, Survived, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, Mr. Bruce Ismay, Male, 29, 0, 0, 15167, 51.0, C85, S
2, 1, Mr. William, Male, 34, 1, 0, 17463, 51.0, C123, S
3, 1, Mr. John Bradley, Male, 22, 1, 0, 17599, 51.0, C123, S
4, 1, Mr. John Bradley, Male, 22, 1, 0, 17599, 51.0, C123, S
5, 1, Mr. John Bradley, Male, 22, 1, 0, 17599, 51.0, C123, S

```

```

1 # Import pandas as pd
2 # Import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # 1. Display the first 5 rows of the dataset
8 print(data.head())
9
10 # 2. Display the last 5 rows of the dataset
11 print(data.tail())
12
13 # 3. Get the shape of the dataset
14 print(data.shape)
15
16 # 4. Get a summary of the dataset
17 print(data.info())

```

Test Case 1: PassengerId Survived Parch Fare Cabin Embarked

Course

mtioo.cocdetantra.com/secure/course/pp?courseid=695c4863a95214c8b7d5e#focmmtz/598154810a5b96214c5b7d58/695c4863a95214c8b7d5e/695c4863a95214c8b7d5e/1028471d94cad0a6

CodeTANTRA Home Learn Anywhere

2023/10/40196@mtioo.co.in Support Logout

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median age of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Braund, Mr. Owen	male	22	1	0	71	5.12	53	S
2	1	1	Mr. Mr. Bradley (Llewellyn Thomas)	male	29	1	0	10540	12.3675	C85	C
3	1	3	Mr. Kelly, Mrs. Ellen	female	33	0	0	9370	7.25	S	S
4	1	1	Mr. Kelly, Mrs. Jacques Heath (Lily May)	female	35	1	0	113801	53.1	C125	S
5	0	3	Mr. Allen, Mr. William Henry	male	35	0	0	274310	53.1	S	S
6	0	3	Mr. Brown, Mr. Samuel	male	36	0	0	155021	51.66	S	S

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, ParCh, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Mr. Braund, Mr. Owen", male, 22, 1, 0, 71, 5.12, 53, S
2, 1, 1, "Mr. Mr. Bradley (Llewellyn Thomas)", male, 29, 1, 0, 10540, 12.3675, C85, C
3, 1, 3, "Mr. Kelly, Mrs. Ellen", female, 33, 0, 0, 9370, 7.25, S, S
4, 1, 1, "Mr. Kelly, Mrs. Jacques Heath (Lily May)", female, 35, 1, 0, 113801, 53.1, C125, S
5, 0, 3, "Mr. Allen, Mr. William Henry", male, 35, 0, 0, 274310, 53.1, S, S
6, 0, 3, "Mr. Brown, Mr. Samuel", male, 36, 0, 0, 155021, 51.66, S, S
```

ItemsDel...

```
1 # Load the Titanic dataset
2 data = pd.read_csv('Titanic-Dataset.csv')
3 data['FamilySize'] = data['SibSp'] + data['ParCh']
4
5
6
7
8
9
10
11
12 # Load the Titanic dataset
13 data = pd.read_csv('Titanic-Dataset.csv')
14 data['FamilySize'] = data['SibSp'] + data['ParCh']
15
16
17 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
18
```

Average time: 0.140 s, Maximum time: 0.140 s, 1 out of 1 shown test case(s) passed

Test case 1: 100%, Debug, Test cases

Expected output: 20, 6032754702882, 17, 4542, 5, 491, 1, 216, 2, 194, Name: Pclass, dtype: int64

Actual output: 20, 6032754702882, 17, 4542, 5, 491, 1, 216, 2, 194, Name: Pclass, dtype: int64

Course

mtioo.cocdetantra.com/secure/course/pp?courseid=695c4863a95214c8b7d5e#focmmtz/598154810a5b96214c5b7d58/695c4863a95214c8b7d5e/695c4863a95214c8b7d5e/1028471d94cad0a6

CodeTANTRA Home Learn Anywhere

2023/10/40196@mtioo.co.in Support Logout

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarked location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Braund, Mr. Owen	male	22	1	0	71	5.12	53	S
2	1	1	Mr. Mr. Bradley (Llewellyn Thomas)	male	29	1	0	10540	12.3675	C85	C
3	1	3	Mr. Kelly, Mrs. Ellen	female	33	0	0	9370	7.25	S	S
4	1	1	Mr. Kelly, Mrs. Jacques Heath (Lily May)	female	35	1	0	113801	53.1	C125	S
5	0	3	Mr. Allen, Mr. William Henry	male	35	0	0	274310	53.1	S	S
6	0	3	Mr. Brown, Mr. Samuel	male	36	0	0	155021	51.66	S	S

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, ParCh, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Mr. Braund, Mr. Owen", male, 22, 1, 0, 71, 5.12, 53, S
2, 1, 1, "Mr. Mr. Bradley (Llewellyn Thomas)", male, 29, 1, 0, 10540, 12.3675, C85, C
3, 1, 3, "Mr. Kelly, Mrs. Ellen", female, 33, 0, 0, 9370, 7.25, S, S
4, 1, 1, "Mr. Kelly, Mrs. Jacques Heath (Lily May)", female, 35, 1, 0, 113801, 53.1, C125, S
5, 0, 3, "Mr. Allen, Mr. William Henry", male, 35, 0, 0, 274310, 53.1, S, S
6, 0, 3, "Mr. Brown, Mr. Samuel", male, 36, 0, 0, 155021, 51.66, S, S
```

ItemsDel...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['ParCh']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10
11 # 1. Calculate the survival rate by class
12 print(data.groupby('Pclass')['Survived'].mean())
13
14 # 2. Calculate the survival rate by embarked location
15 print(data.groupby('Embarked_S')['Survived'].mean())
```

Average time: 0.105 s, Maximum time: 0.105 s, 1 out of 1 shown test case(s) passed

Test case 1: 100%, Debug, Test cases

Expected output: Pclass, 1, 0.62028, 2, 0.47826, 3, 0.24716, Name: Survived, dtype: float64

Actual output: Pclass, 1, 0.62028, 2, 0.47826, 3, 0.24716, Name: Survived, dtype: float64

